

← Main menu

Build deterministic Virtual Agent enhanced with data stores

Course · 1 hour 30 minutes · < 1% complete

Course overview

Challenge Lab

Build Deterministic Virtual Agents With Your Own Data Stores: Challenge Lab

Your Next Steps

Course Badge

Course Survey

Course > Build deterministic Virtual Agent enhanced with data stores >

Quick tip: Review the prerequisites before you run the lab

End Lab 00:30:54

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. [Learn more.](#)

Open Google Cloud Console

Username

student-02-9d1c65496ec9f

Password

R78yP39UfwUC

Project ID

qwiklabs-gcp-04-2c7c467f

Build Deterministic Virtual Agents With Your Own Data Stores: Challenge Lab

Lab 1 hour 30 minutes No cost Intermediate

★★★★☆ Rate Lab

This lab may incorporate AI tools to support your learning.

Lab instructions and tasks 80/100

Challenge Lab Overview

Objective

Setup

Challenge Scenario

Your challenge

Task 1. Build a deterministic bot using Conversational Agents

Task 2. Deploy a Chatbot built using AI Applications

Congratulations!

< Previous

Challenge Lab Overview

or following step-by-step instructions, you are given a common business scenario and a set of tasks - you figure out how to complete them on your own! An automated scoring system (shown on this page) provides feedback on whether you have completed your tasks correctly.

When you take a Challenge Lab, you will not be taught Google Cloud concepts. You will need to use your skills to assess how to build the solution to the challenge presented. This lab is only recommended for students who have those skills. Are you up for the challenge?

Objective

The objective of this challenge lab is to demonstrate your proficiency in deploying chat applications and customizing chatbots through Conversational Agents and AI Applications. Additionally, emphasis is placed on constructing a tailored Q&A system utilizing data stores. In this lab, you will be ready to deploy conversational solutions for

Next >

Setup

Qwiklabs setup

For each lab, you get a new Google Cloud project and set of resources for a fixed time at no cost.

1. Make sure you signed into Qwiklabs using an incognito window.

2. Note the lab's access time (for example, 02:00:00) and make sure you can finish in that time block.

There is no pause feature. You can restart if needed, but you have to start at the beginning.

3. When ready, click START LAB

4. Note your lab credentials. You will use them to sign in to the Google Cloud

Open Google Console

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. [Learn more.](#)

Username

student-02-9d1c65496ec9f

Console. `google@d8/b52b_student@qwiklabs.n`

Password `TG959yrKDX`

GCP Project ID `qwiklabs-gcp-0855e773352d3560`

5. Click **Open Google Console**.

6. Click **Use another account** and copy/paste credentials for **this** lab into the prompts.

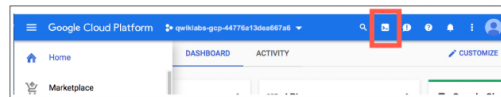
If you use other credentials, you'll get errors or **incur charges**.

7. Accept the terms and skip the recovery resource page.

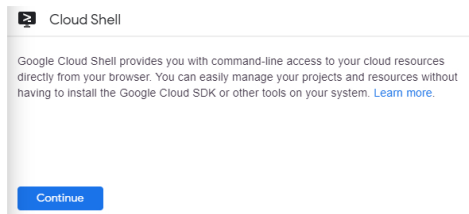
Do not click **End Lab** unless you are finished with the lab or want to restart it. This clears your work and removes the project.

Activate Cloud Shell

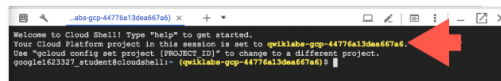
Cloud Shell is a virtual machine that is loaded with development tools. It offers a persistent 5GB home directory and runs on the Google Cloud. Cloud Shell provides command-line access to your Google Cloud resources.



Click **Continue**.



It takes a few moments to provision and connect to the environment. When you are connected, you are already authenticated, and the project is set to your **PROJECT ID**.



`gcloud` is the command-line tool for Google Cloud. It comes pre-installed on Cloud Shell and supports tab-completion.

You can list the active account name with this command:

```
gcloud auth list
```

(Output)

```
Credentialed accounts:
- <myaccount>@<mydomain>.com (active)
```

(Example output)

You can list the project ID with this command:

```
gcloud config list project
```

(Output)

```
[core]
project = <project_ID>
```

(Example output)

```
[core]
project = qwiklabs-gcp-44776a13dea667a6
```

For full documentation of `gcloud` see the [gcloud command-line tool overview](#).

Challenge Scenario



Here is a company overview as provided on Cymbal Shops's website.

Cymbal Shops is an American retail chain headquartered in Minneapolis that sells homewares, electronics, and clothing.

Founded in 1974, Cymbal Shops started out as Cymbal Air, selling AC systems manufactured by Cymbal Group in Minnesota and neighboring states. The company quickly expanded into domestic merchandise in order to satisfy the need for quality products at an affordable price in the midst of a recession.

Cymbal Shops' broad product assortment was once a benefit - capturing planned purchases and last minute splurges. In recent years, however, the company has struggled to adapt to the acceleration in e-commerce. Digitally native companies are on the rise and Cymbal Shops must implement significant changes in order to keep pace and maintain relevance. Today, Cymbal Shops operates 714 stores across North America and reported \$15 billion in revenue in 2019. They currently employ 80,400 employees across the United States and Canada.

Cymbal Shops is a digitally transforming legacy retailer.

Your challenge

You work as an engineer at Cymbal Shops, a retail chain company renowned for its innovative solutions. Cymbal Shops is embarking on a significant project that involves developing robust search functionality and a virtual assistant for a new application. Your task is to design a Proof of Concept (POC) solution that seamlessly creates and deploys Virtual Agents using Conversational Agents and AI Applications and focuses on generating generative capabilities using AI Applications and Gemini.

Cymbal Shops' new application demands a powerful and efficient search feature, aiming to catalog and retrieve various data types, including text documents, images, and structured data using AI Applications. Additionally, the company aims to create a virtual assistant using AI Applications to assist customers with queries about products and devices available in the Google Store. The virtual assistant will be built and configured using AI Applications.

Therefore, in this lab, you should:

1. Build and deploy a deterministic chatbot using Conversational Agents.
2. Add generative capabilities using AI Applications and Gemini.

Task 1. Build a deterministic bot using Conversational Agents

In this section, use Conversational Agents to create a deterministic bot using a sample

1. Enable [AI Applications API](#).
2. In [Conversational Agents console](#) create an **Agent** named **Store Agent** in `global` region.

- Greet you
- Create an appointment
- Ask for a location
- Ask for a date
- Validate for open slots 9 am to 11 am
- Ask for a name for the reservation
- Confirm the appointment

6. Once the agent has collected the necessary information, create a routing condition that routes the end user to another page named **Confirm Appointment**, based on the following routing condition:

Condition rules: `$page.params.status = "FINAL"`

Great \$session.params.name. You are all set for next \$session.params.date at \$session.params.time in our \$session.params.location

9. Add two distinct routes for the Confirm Appointment page, namely `confirmation.yes` and `confirmation.no`, as mentioned below

10. Additionally, set the parameter presets for the `confirmation.no` intent to the following:

11. Set the transition of `confirmation.no` intent to a new page named `Appointment Information`.

Finally, the agent should respond with the following when the **Confirm Appointment** page is active.

Great your appointment is booked!



Output:

Equipment: Draft

Page: Start Page

What day of the week would you like the appointment for?

Tomorrow

What time is convenient for you? Our available time slots are mornings, from 9:00 AM to

10 am

What name should I put in the reservation?

Andy

next 2024-06-05 at 10:00:00 in our Austin

Great your appointment is booked!

Click **Check my progress** to verify the objective.



Build a deterministic bot using Conversational Agents

Check my progress

Assessment Completed!

Task 2. Deploy a Chatbot

In this section, you will incorporate a data store into the bot, enabling customers to pose questions to the **Store Agent**. You will utilize a Cloud Storage location to establish the data store.

Hence, in this task, you will deploy a chatbot constructed using AI Applications with the specified configurations.

1. For **Start Page**, connect to **Data stores** using the **Add state handler** option.
2. To add data stores, create a Vertex AI Search and Conversation app of type **Chat** in the **Global** region, linked to the agent. Configure this app with a new data store named **Google Store**, directing to unstructured data hosted in the following Cloud Storage bucket: `cloud-samples-data/dialogflow-cx/google-store`.

Note: Wait until the data is imported successfully.

3. In the **Conversational Agents** console, link the agent to the data store named **Google Store** using the **Add state handler** feature from the **Start Page**.

4. **Start Page** is the **Default Welcome** intent used by default.

5. Test the agent by asking the following questions:

- Hello
- How long does the battery in the Pixel 7 Pro last?
- Is the Pixel Watch water resistant?
- Can I display my Google Photos on a Nest Hub?

Output:

Environment: Draft
Flow: Default Start Flow
Page: Start Page



Good day! What can I do for you today?



How long does the battery in the Pixel 7 Pro last?



The Pixel 7 Pro has an estimated battery life of up to 72 hours when using Extreme Battery Saver mode, which disables various features including 5G connectivity. The average battery life during testing was approximately 31 hours.



Is the Pixel Watch water resistant?



Pixel Watch is water resistant up to 50 meters (5 ATM). You can swim in the ocean, play in the pool, take a shower, and more without worrying. However, it is not waterproof.

Click **Check my progress** to verify the objective.



Deploy a Chatbot built using AI Applications

Check my progress

Please create a generative chat application using datastore 'Google Store' as specified in the lab instructions.

Congratulations!

You have successfully navigated the complexities of building deterministic virtual agents with your own data stores on Google Cloud in this hands-on lab. By developing a basic, deterministic chatbot using Conversational Agents that can understand and respond to user requests according to a defined script, you created a chatbot that can answer user questions about products in the Google Store by accessing information from Cloud Storage.

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